Strava Fitness Data Analytics

Tableau & SQL Insights Report

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Overview

For this project, cleaned FitBit activity, sleep, and hourly datasets were loaded into a MySQL database, queried with SQL to answer key behavioural questions, and connected to Tableau to build an interactive, multi-view dashboard. The goal was to understand user activity patterns across the day and week, the relationship between activity and calories burned, and how engagement varies across the user base.

The dashboard is organised into three linked views, each highlighting a different angle on user behaviour:

- **Average Steps by Day of Week** — shows how activity shifts across the week.
- **Average Steps by Hour** — shows the intraday rhythm of activity.
- **Steps vs. Calories** — shows the relationship between daily steps and calories burned, coloured by user.

Insights

1. Weekly Activity Pattern

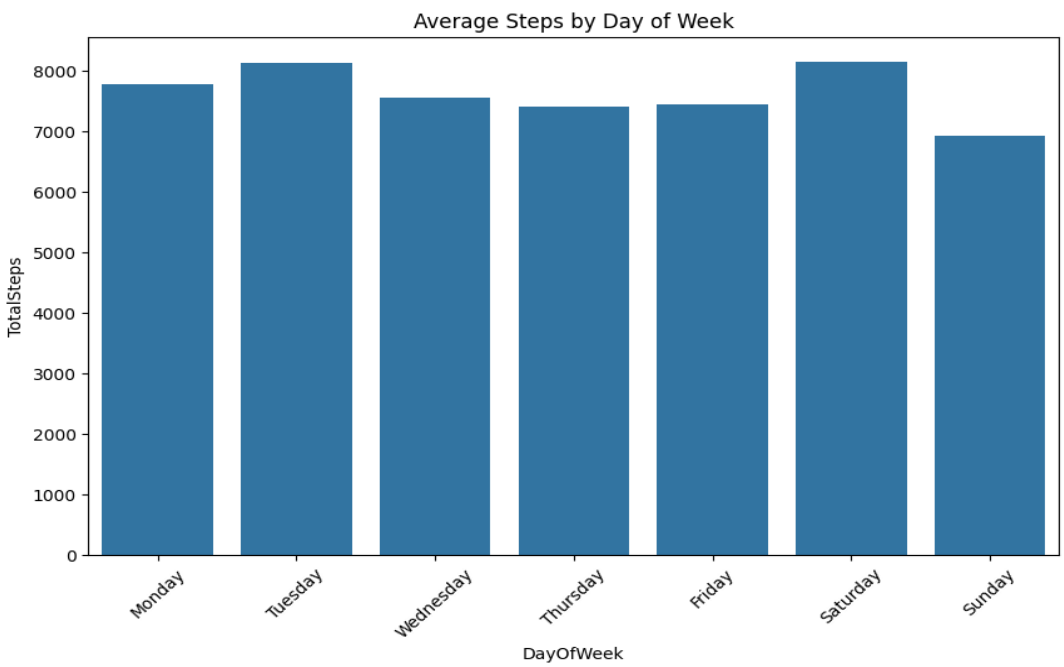

Figure 1. Average steps by day of week (Colab / seaborn output).

- Activity dips midweek and rises toward the weekend, with Tuesday and Saturday standing out as the most active days.
- Sunday shows the lowest average step count of the week, suggesting a consistent rest-day pattern across users.

• The spread between the highest and lowest day is moderate rather than extreme, indicating fairly stable weekly habits overall.

2. Hourly Activity Pattern

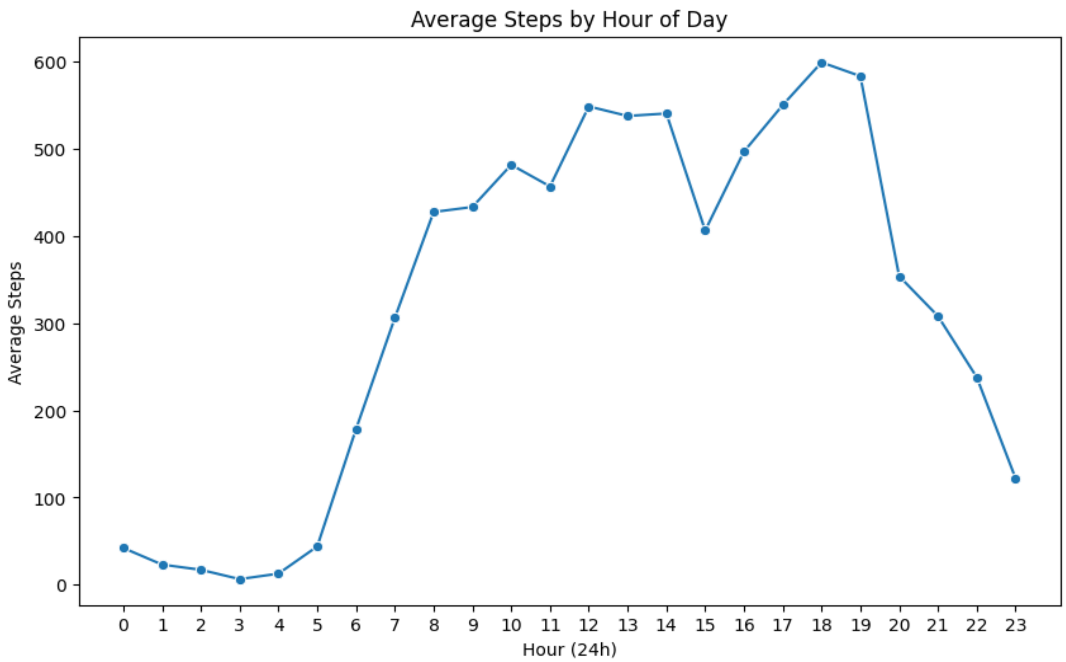

Figure 2. Average steps by hour of day.

• Activity is minimal overnight (roughly 12am–5am), as expected.

• Steps climb steadily through the morning and stay elevated through the afternoon and early evening.

• The clearest peak falls around 6–7pm, consistent with an after-work or after-school activity window, before tapering off toward midnight.

3. Steps vs. Calories

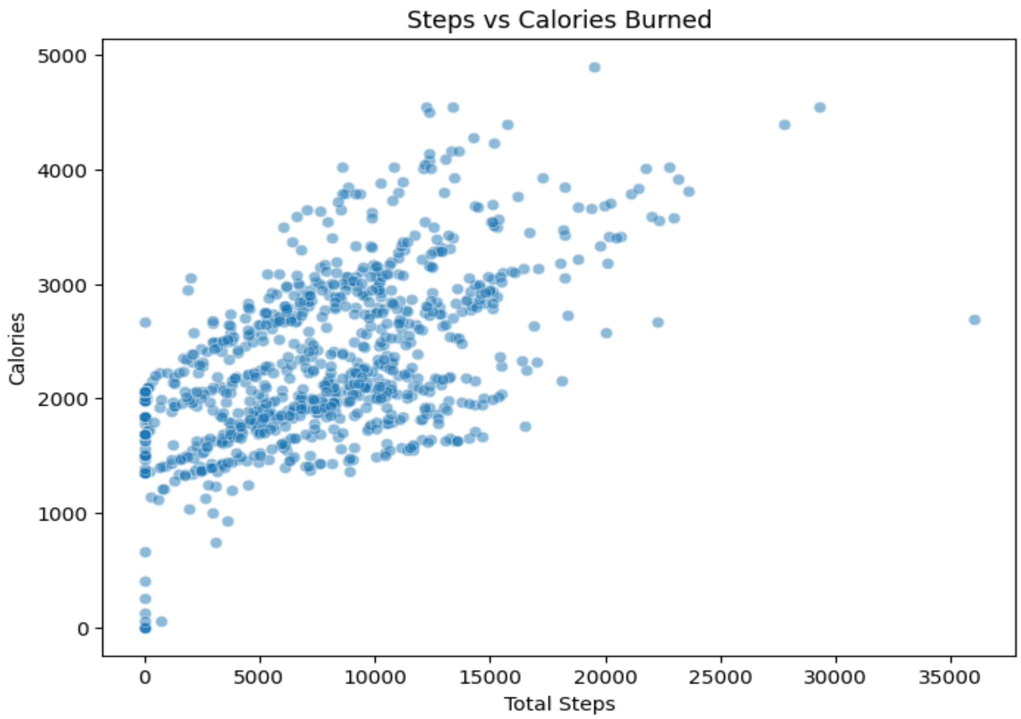

Figure 3. Total steps vs. calories burned, per user-day.

- There is a clear positive relationship between daily steps and calories burned across users.
- The relationship is not perfectly linear — some users burn more calories at a given step count than others, likely reflecting differences in workout intensity, body composition, or activity type not captured by step count alone.

4. Steps vs. Sleep

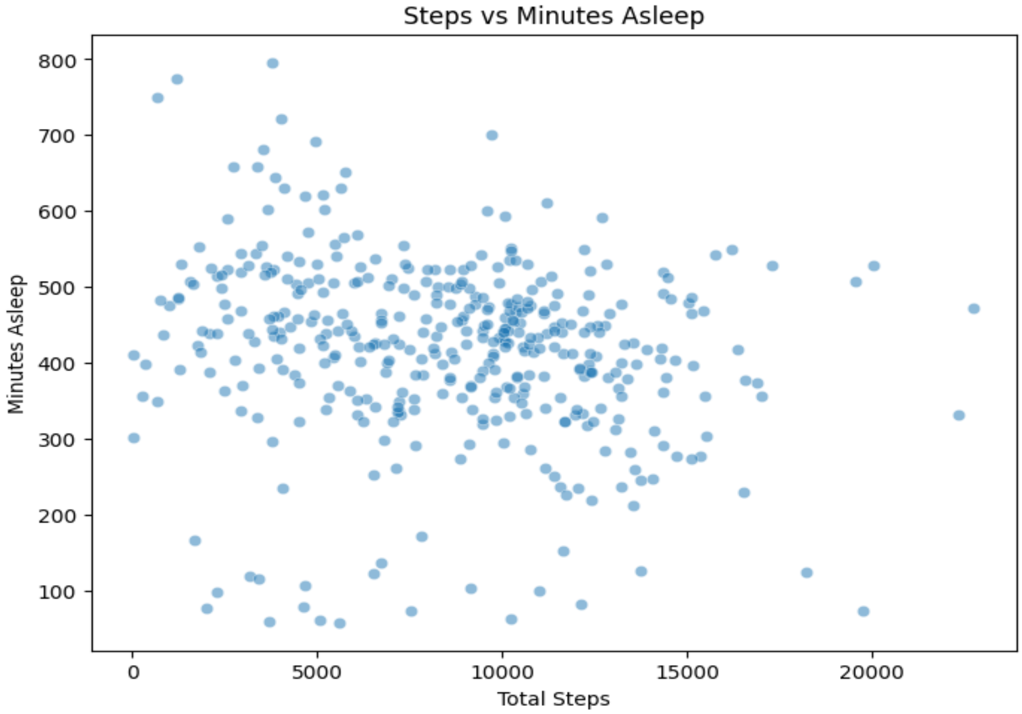

Figure 4. Total steps vs. minutes asleep, per user-day.

• Fewer users logged sleep data than activity data, a common gap in self-reported tracker datasets.

• This limited the depth of sleep-specific analysis; sleep is included in the underlying data but is not a core dashboard view for this reason.

5. Activity Composition

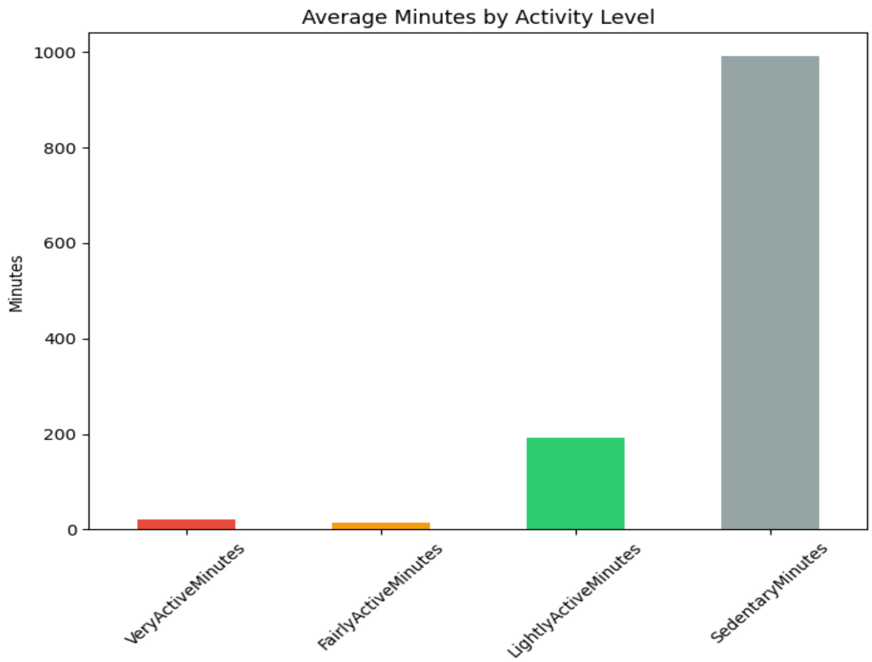

Figure 5. Average minutes spent per activity intensity level.

6. User Segmentation

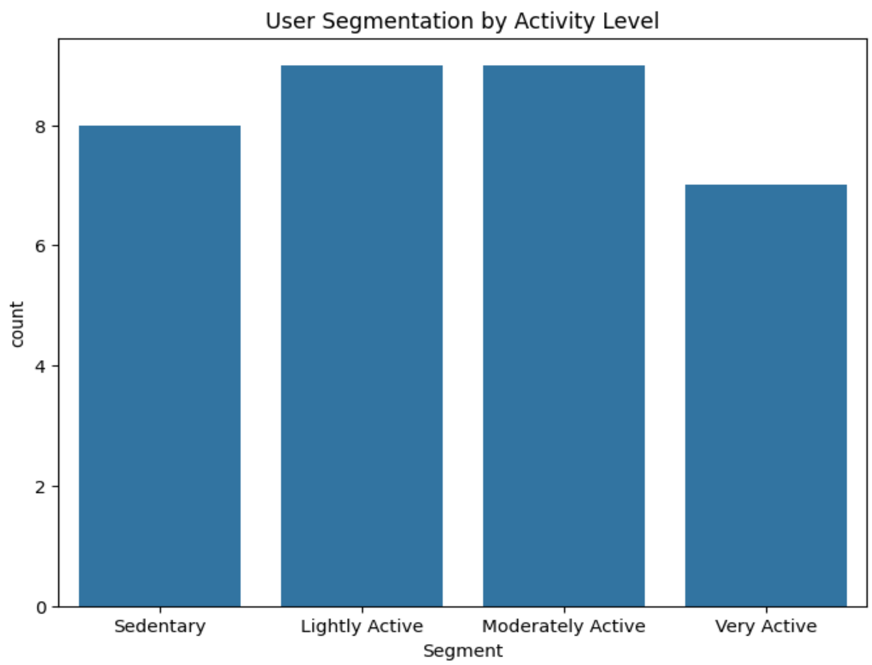

Figure 6. Users segmented by average daily steps.

• Users were segmented by average daily steps into four bands: Sedentary (under 5,000), Lightly Active (5,000–7,499), Moderately Active (7,500–9,999), and Very Active (10,000+).

• Activity levels vary widely across the user base, with no single segment dominating — reflecting a realistic mix of habitual and occasional exercisers.

Actionable Insights

Midweek & weekend engagement	Tuesday and Saturday already peak — campaigns (challenges, reminders) may land better timed for lower-activity days instead, particularly Sunday.
Evening engagement window	The 6–7pm peak suggests this is when users are most receptive to prompts, workout suggestions, or social features.
Sleep data collection	Encouraging more consistent sleep logging would unlock deeper analysis of the activity–sleep relationship.
Segment-aware coaching	The spread across activity segments suggests personalised goals (e.g. step targets scaled to a user's current segment) could be more effective than a single blanket target.

Conclusion

By connecting a cleaned MySQL database to Tableau and building a linked, interactive dashboard, this project turns raw FitBit tracker exports into a clear picture of when and how users are active. The weekly and hourly patterns, along with the steps–calories relationship, offer a foundation for targeted engagement and personalised activity goals.

Dashboard & Repository

Live interactive dashboard:
public.tableau.com/app/profile/shreya.jain4118/viz/Book1_17874683186530/Dashboard1

Dataset: FitBit Fitness Tracker Data (Kaggle, CC0) — kaggle.com/datasets/arashnic/fitbit

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